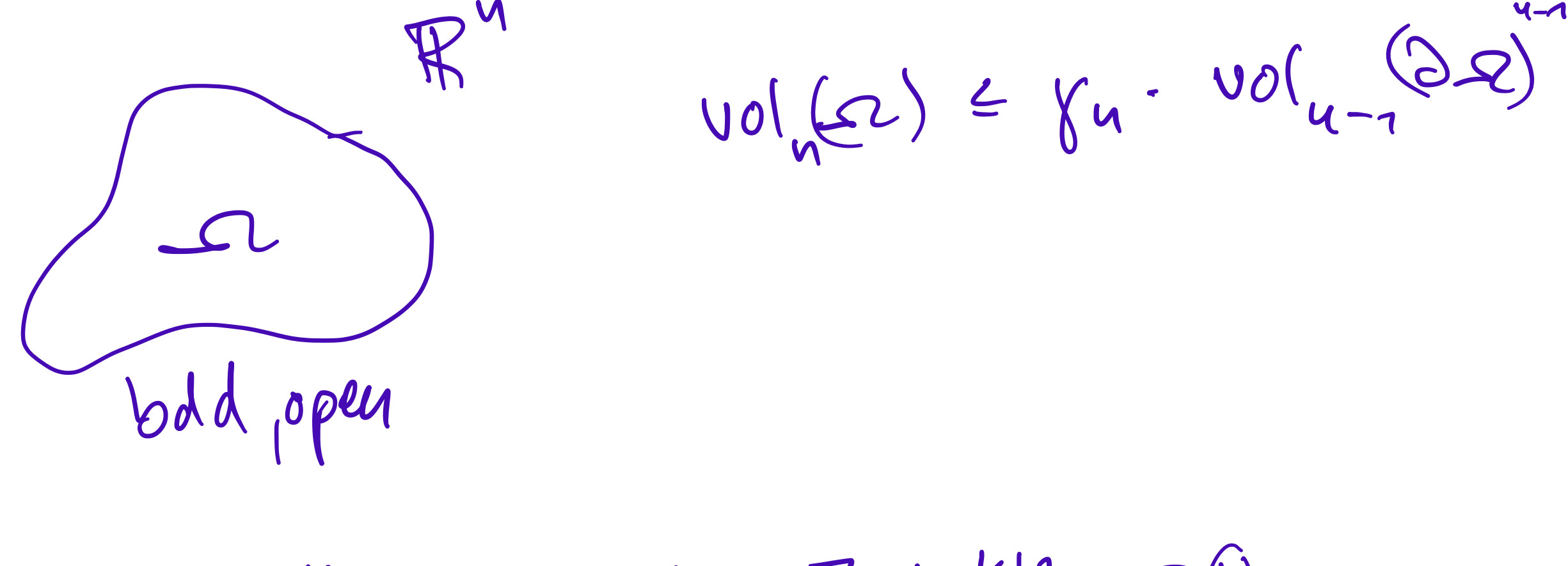


An isoperimetric gap in NPC (joint C. Druţu, U. Lang, P. Papasoglu)



$$Z^k \subset \mathbb{R}^n, \partial Z \neq \emptyset, \exists Y^{k+1} \subset \mathbb{R}^n, \partial Y = Z$$

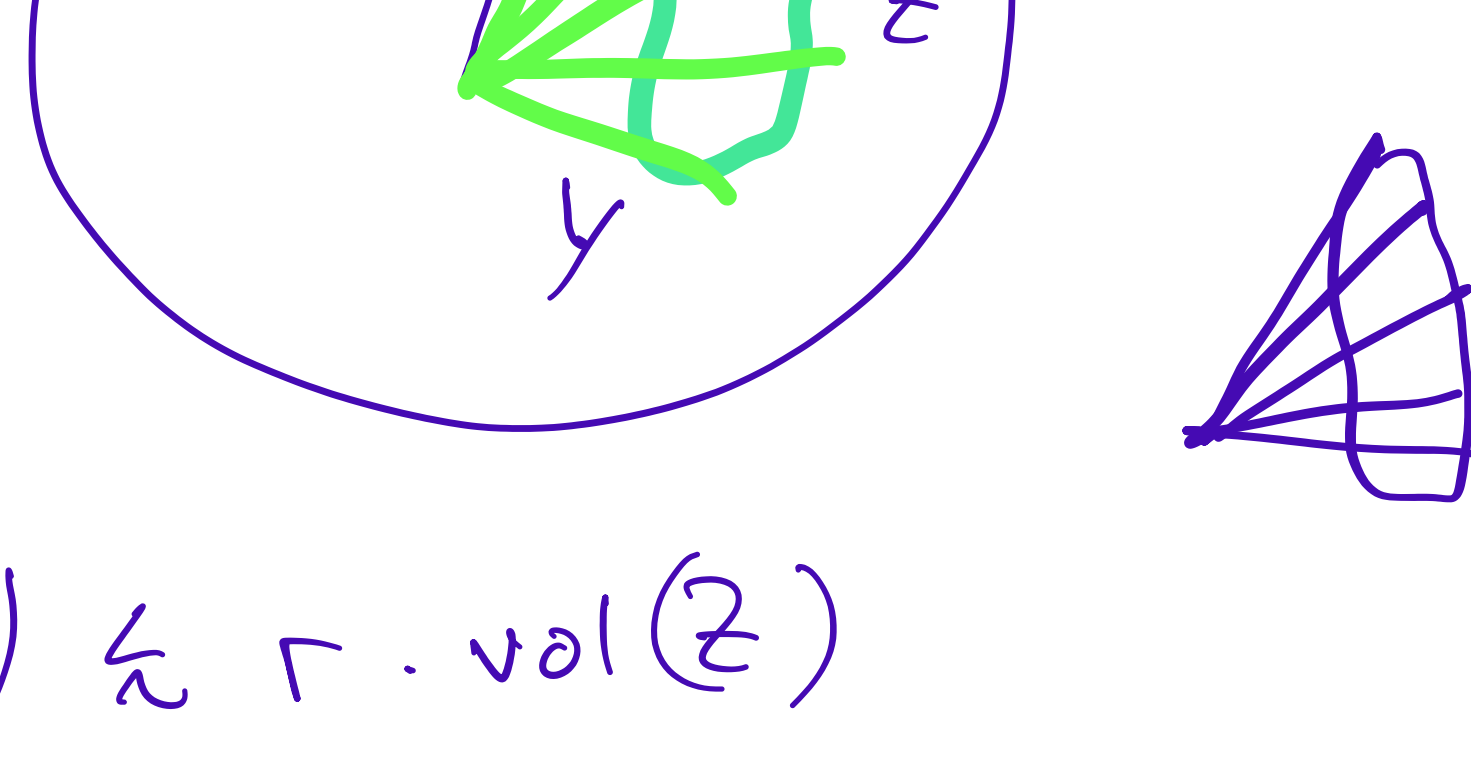
$$\boxed{\text{vol}_{k+1}(Y) \leq C \cdot \text{vol}_k(Z)^{\frac{k+1}{k}}} \quad (E1)$$

- Federer - Fleming (1959) $C = C(n)$
- Michael - Simon (1973) $C = C(k)$
- Alluening (1986) $C = \gamma_{k+1}$

$$\gamma_{k+1} = \frac{\text{vol } \mathbb{B}^{k+1}}{\text{vol } (\partial \mathbb{B}^{k+1})^{\frac{k+1}{k}}}$$

Wenger's algorithm (2008)

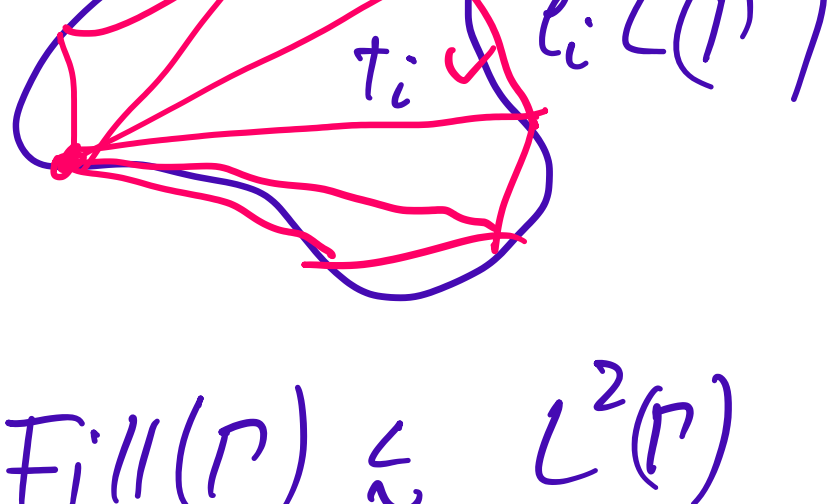
Cone inequ.



$$\text{vol}(Y) \lesssim r \cdot \text{vol}(Z)$$

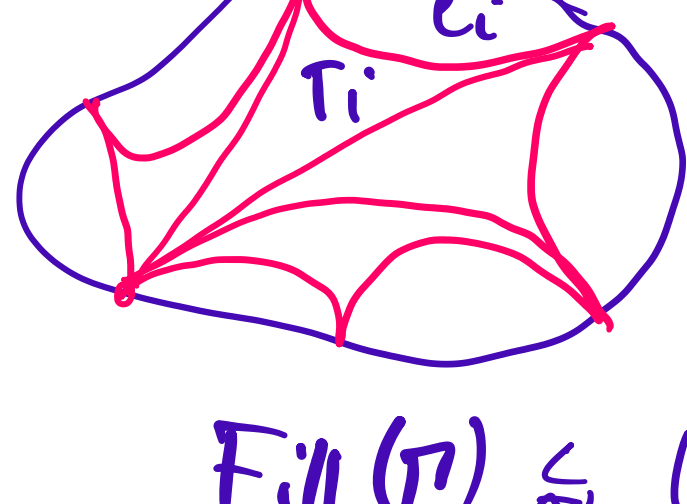
Isop. inequ. in NPC

$K \leq 0$



$$\text{Fill}(r) \leq L^2(r)$$

$K \leq -1$



$$\text{Fill}(r) \leq L(r)$$

Reshetnyak: $X \text{ CAT}(0), r \subset X$

$$\text{Fill}(r) \leq \frac{1}{4\pi} \cdot L(r)^2$$

Wenger's gap theorem:

$X \ni \varepsilon > 0, L_0 > 0$ st

$$\text{Fill}(r) \leq \frac{1-\varepsilon}{4\pi} \cdot L(r)^2, L(r) \geq L_0$$

$\Rightarrow X$ Gromov hyperbolic

$\Leftrightarrow X_w$ is a tree

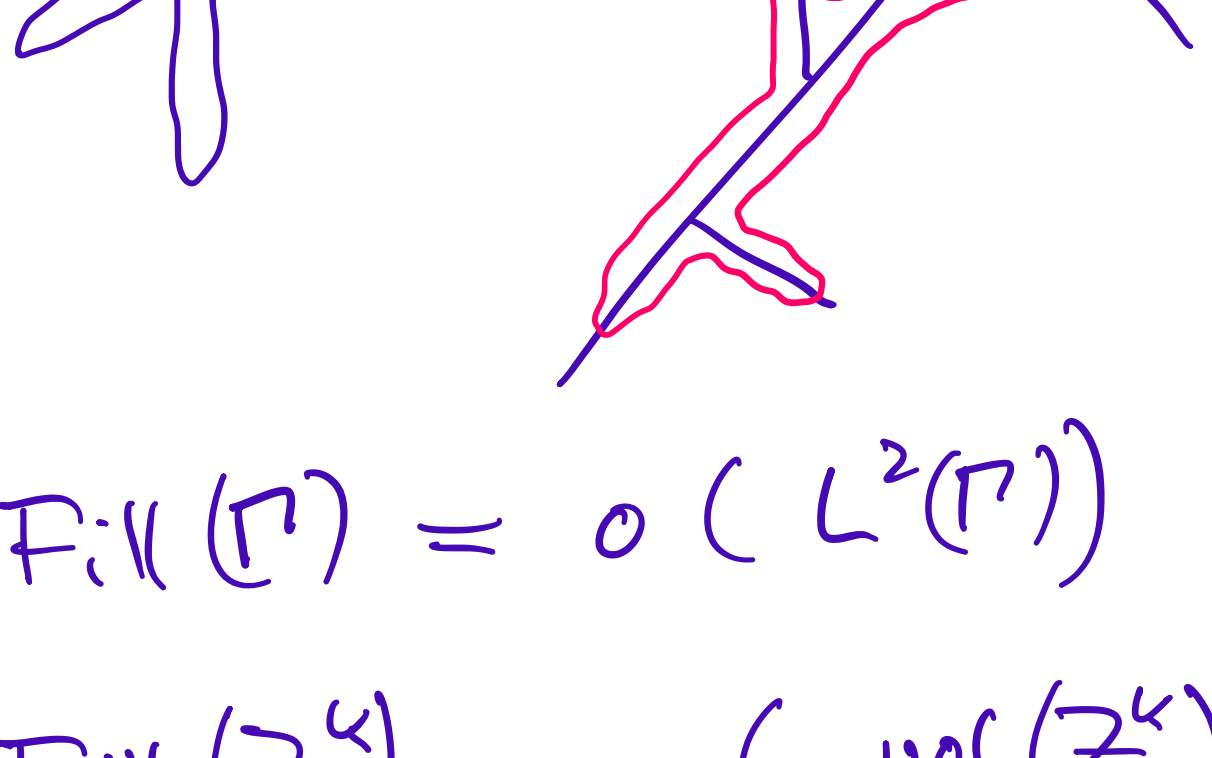
$\xRightarrow{\text{CAT}(0)} \dim X_w = 1$

$$\Rightarrow \text{Fill}(r) \leq C \cdot L(r)$$

Gromov asymp. rk $(X) = \sup \dim X_w$

$(X \text{ cocomp}) \text{ asrk} = \max \{k \in \mathbb{N} \mid \mathbb{R}^k \xrightarrow{\text{isom}} X\}$

$\lambda \cdot X$ close X_w tree



$$\text{Fill}(r) = o(L^2(r))$$

$$\text{Fill}(Z^k) = o(\text{vol}(Z^k)^{\frac{k+1}{k}})$$

Conjecture: If $\text{asrk}(X) = k$

$$\text{Fill}(Z^k) \leq C \cdot \text{vol}(Z^k)$$

True: sym. spaces

Let X be $\text{CAT}(0)$, loc. ccomp

$f: S^2 \rightarrow X$ lip.

$$\text{Fill}(f) = \inf \{ \text{vol}(\hat{f}) \mid \hat{f}: \mathbb{B}^3 \rightarrow X \text{ lip, st. } \hat{f}|_{S^2} = f \}$$

Thm 1: TFAE

i) $\exists C < \frac{1}{64\pi}, A_0 > 0$:

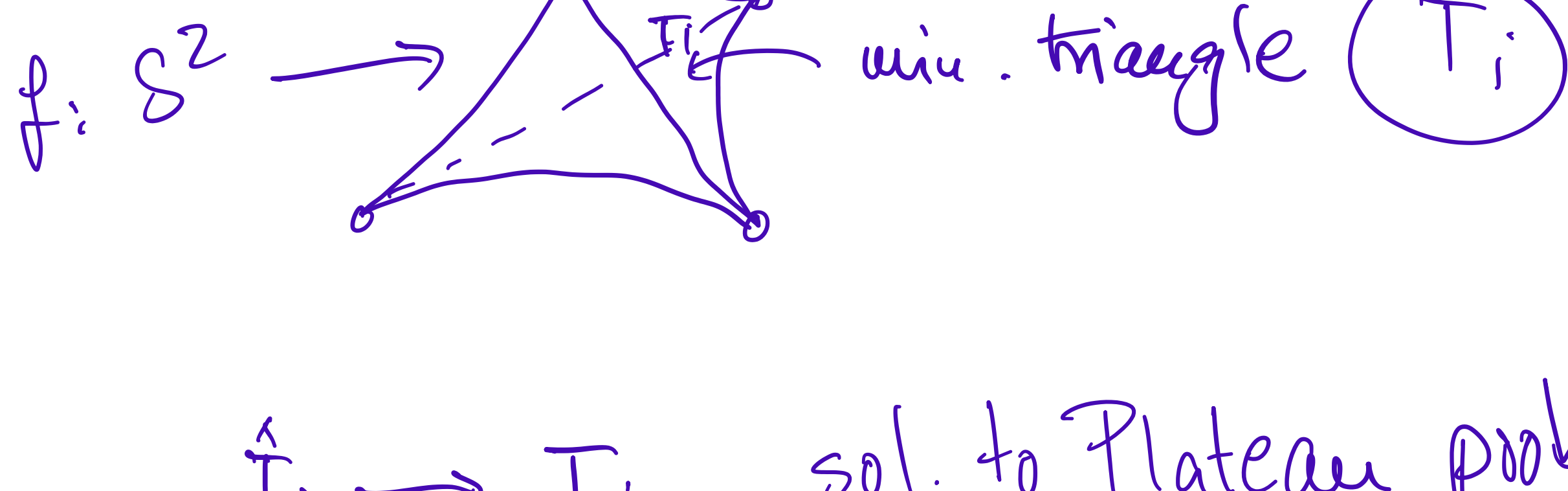
$$\text{Fill}(f) \leq C \cdot A(f)^{\frac{3}{2}}, A(f) \geq A_0$$

ii) $\forall \delta \in (0, 1/2) \exists C_\delta > 0$:

$$\text{Fill}(f) \leq C_\delta \cdot A(f)^{1+\delta}$$

iii) $\text{asrk}(X) \leq 2$

Minimal tetrahedra



$u: \hat{T}_i \rightarrow T_i$ sol. to Plateau problem



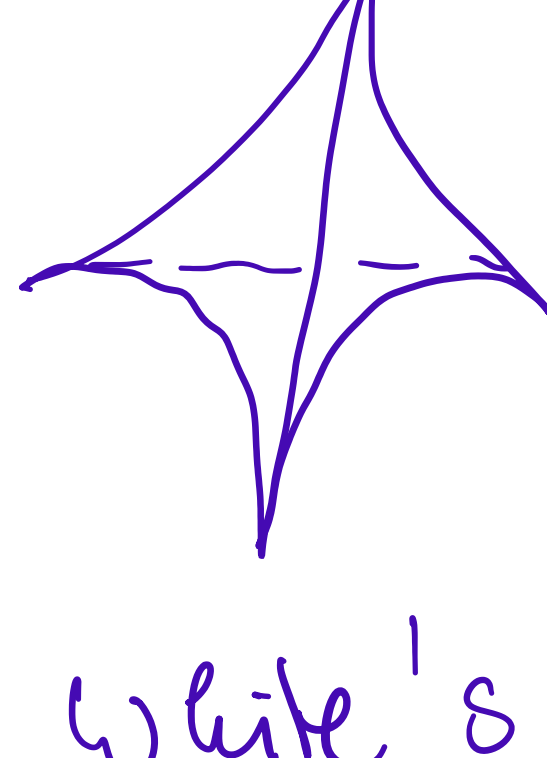
Thm 2: If $\text{asrk}(X) \leq 2$, then

$$\text{Fill}(T) \leq C \cdot A(T)$$

$$1) A(T_i \cap B_r(p)) \leq \frac{3\pi}{2} \cdot r^2 \quad \forall p \quad \forall r > 0$$

$$2) \text{Fillrad}(T) \leq C \quad (\text{Goldkirsch - Lang})$$

Kleiner - Lang



3) White's trick

Proof: Thm 2 $\Rightarrow$ Thm 1 i) δ -isop.

Ind. on scale

Ind base (E1) $\text{Fill}(S) \leq C \cdot A(S)^{3/2} \leq C \cdot A(S)^{1+\delta}$ if $A(S) < 1$

Wenger's algo.

2nd step: $f: S^2 \rightarrow X, L$ -lip, $A(f) \sim L^2$

$$s \in (0, 1) \quad T_n^{(2)} \sim \frac{1}{s^2}, L(\partial \Delta_i) \sim s$$

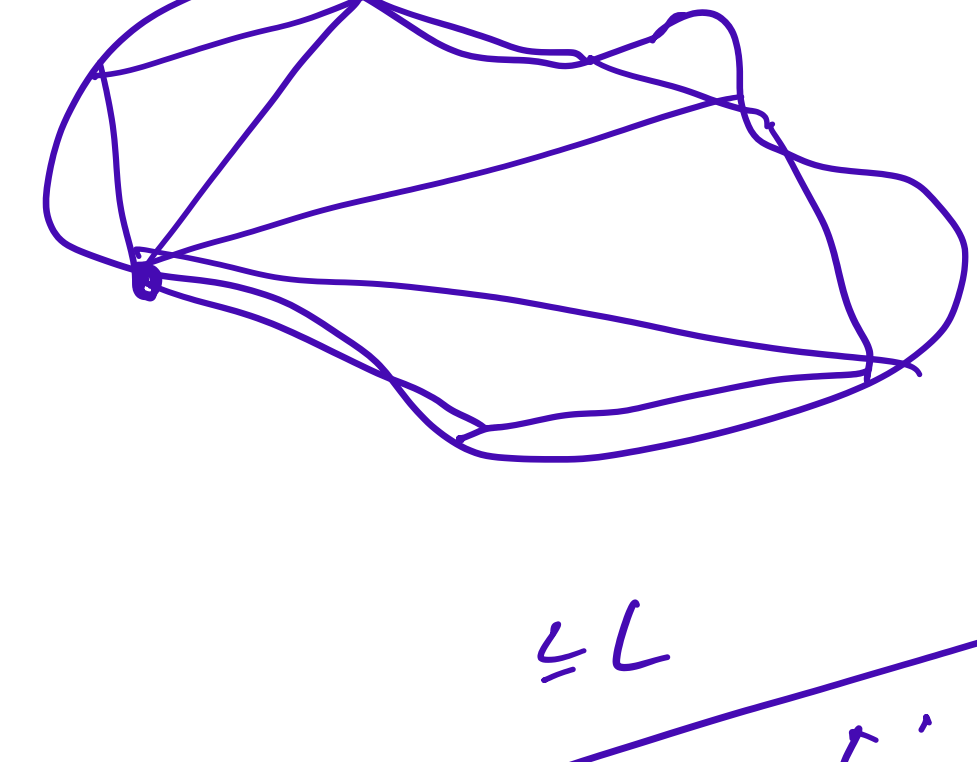
$$L^2 S^2 \quad f(\Delta_i) \leq Ls \quad \sigma_i \quad A(\sigma_i) \leq L^2 s^2 < A(f)$$

$$\Rightarrow \text{Fill}(\sigma_i) \leq A(\sigma_i)^{1+\delta}$$

$h: S^2 \times [0, 1] \rightarrow X$

$$\text{vol}(h) \leq \frac{1}{s^2} \cdot (L^2 s^2)^{1+\delta} = L^{2+2\delta} \cdot s^{2\delta}$$

$f \sim \frac{1}{q} f'$ pw min.



$$\text{Fill}(\sigma_i) \leq L^2 s$$

$$\text{Fill}(f) \leq \frac{1}{s^2} \cdot L^2 s = \frac{L^2}{s}$$

$$\text{Fill}(f) \leq \text{vol}(h) + \text{Fill}(f') \leq L^{2+2\delta} \cdot s^{2\delta} + \frac{L^2}{s}$$

$$\uparrow \quad L^{2+2\delta-2\delta^2} + L^{2+\delta} \leq L^{2+2\delta}$$

choose $s = L^{-\delta}$

